



JARAMOGI OGINGA ODINGA UNIVERSITY OF SCIENCE AND TECHNOLOGY

SCHOOL OF INFORMATICS AND INNOVATIVE SYSTEMS

DEPARTMENT OF INFORMATION SYSTEMS

**UNIVERSITY EXAMINATION FOR THE DEGREE BACHELOR OF
INFORMATION COMMUNICATION AND TECHNOLOGY
YEAR THREE: SEMESTER ONE**

2ND YEAR 2ND SEMESTER 2024/25 ACADEMIC YEAR

MAIN CAMPUS

COURSE CODE: ITB 1302

COURSE TITLE: NETWORK SYSTEM INTEGRATION AND MAINTENANCE

DATE:

TIME:

INSTRUCTIONS

Answer all Questions

Questions One

6. Define **Network Systems Integration** and explain its importance in modern IT infrastructure.
7. Describe three key **network topologies** and their advantages in system design.

8. Explain the role of **Physical Layer protocols** in supporting communication across data networks.
 9. What key steps are involved in **configuring and implementing** a new network system?
 10. Explain the difference between **Network File System (NFS)** and **Network Information Services (NIS)**.
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Questions Two

11. Your company is planning to integrate Linux, Mac OSX, and Windows systems into a single network. Explain the steps you would take to ensure a seamless **network integration**. (10 marks)
12. You are assigned to configure a **Samba server** for file and printer sharing in a small office. List and explain the key configurations needed. (10 marks)
13. A company is experiencing slow network performance. As a **network administrator**, outline the key steps to diagnose and resolve the issue. (10 marks)

Questions Three

14. **System Maintenance and Security**
 - a) Discuss the significance of **network monitoring** in ensuring system security.
 - b) Explain how **system audits** can help optimize network performance. (20 marks)
15. **Project Management in Network Integration**
 - a) Outline the key considerations when planning a **network integration project**.
 - b) Discuss the importance of **documentation** in network system maintenance. (20 marks)

Questions Five

1. **System Analysis and Design**
 - o Explain the concept of **system analysis and design** in the context of network systems.

- Describe the steps involved in gathering customer requirements for a new network system.

2. **Network Administration**

- Define **network administration** and explain its role in maintaining a stable IT infrastructure.
- Identify and explain three key responsibilities of a network administrator.

3. **LDAP and Network Security**

- What is **LDAP**, and how is it used in network authentication?
- Discuss two advantages of using LDAP for identity management in an enterprise network.

4. **File Sharing Protocols**

- Explain the role of **Samba** in a network environment.
- Compare **Samba** with **FTP** and describe a scenario where each would be used.